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SEMICONDUCTOR DIODES

For Students, Professionals and Beyond

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1. Introduction

The **Semiconductor Diode** is a two lead solid-state device that allows current to pass through itself in one defined direction only, thus acting as a sort off one-way valve to current flow. Diodes are commonly made using silicon semiconductor materials, but sometimes other materials, such as germanium, selenium, or gallium arsenide are used.

Diodes are formed by joining together two types of semiconductor material which have had their crystalline structure changed by a process known as doping. This doping process creates an "*n-type material*", and a "*p-type material*".

The resulting semiconductor material created by this fusion of two semiconductor materials creates a positive "p-region" at one end and a negative "n-region" at the other end, thus producing a rectifying device which has a resistivity value somewhere between that of a conductor and an insulator.

But what is a "Semiconductor" material, and what is meant by the terms, n-type and p-type materials. So for us to understand the operation of a *Semiconductor Diode*, we must first look at what makes something either a Conductor or an Insulator.

2. Insulators and Conductors

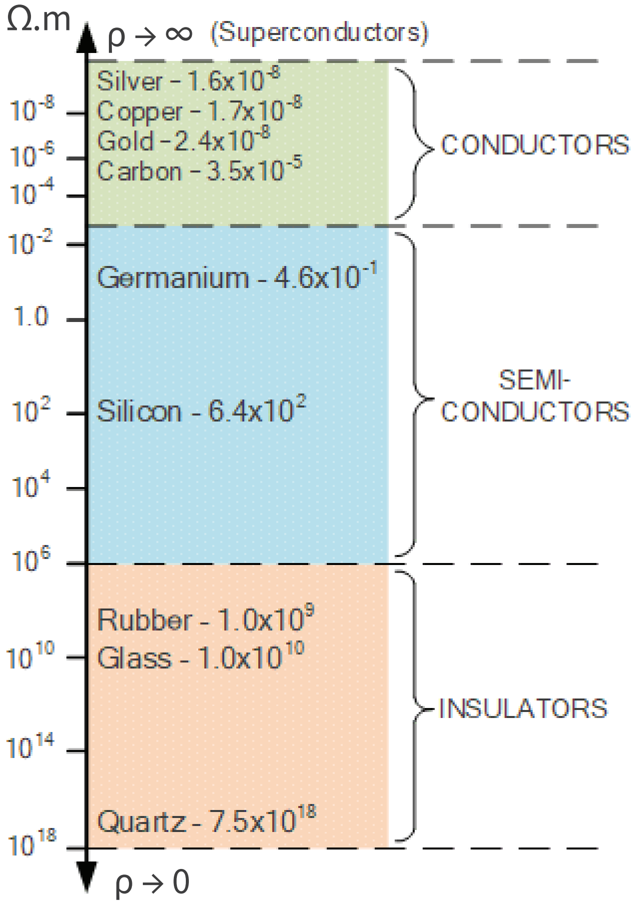
The electrical Resistance of an electrical or electronic component or device is generally defined as being the ratio of the voltage difference across it to the current flowing through it, basic Ohm's Law principals.

The problem with using resistance as a form of measurement is that it depends very much on the physical size of the material being measured as well as the material out of which it is made.

The electrical resistivity of a particular material on the other hand is a measure of how strongly the material opposes the flow of electric current through it. Resistivity is measured in Ohm-metres, (Ω.m).

Thus if the resistivity of various materials is compared, they can be classified into three main groups, Conductors, Insulators and Semiconductors as shown in Figure 1.

Figure 1. Resistivity of Materials



Materials which permit the flow of electrons through themselves are called **Conductors**, such as gold, silver, copper, etc. Conductors have fairly low resistivity.

Materials which block the flow of electrons are called **Insulators**, such as rubber, glass, quartz, etc. Insulators have high resistivity.

Materials whose resistivity falls somewhere between those of a conductor and an insulator are commonly called **Semiconductors**.

The ability of semiconductors to conduct electricity can be greatly improved by replacing or adding certain donor or acceptor atoms to their crystalline structure and silicon is by far the most commonly used semiconductor material for making electronic devices.

Notice that there is a very small margin between the resistivity of the conductors such as silver and gold, compared to a much larger margin for the resistivity of the insulators between glass and quartz.

The difference in resistivity between various materials is due to their ambient temperature as conductive metals are much better absorbers and transmitters of heat than insulators.

3. Semiconductor Basics

Semiconductors materials such as Silicon (Si), Germanium (Ge) and Gallium Arsenide (GaAs), have electrical properties somewhere between those of a "conductor" and an "insulator". Semiconductors consists of atoms that have no net electric charge (having the same number of electrons as protons), so they neither are good conductors of electricity nor are they good insulators (hence their name "semi"-conductors).

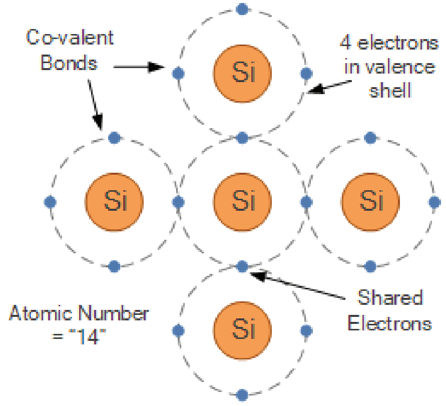
These semiconductor materials have very few "fee electrons" in their valence shell because their atoms are closely grouped together in a tight crystalline pattern called a "crystal lattice".

Semiconductor materials made from silicon are used to produce diodes, transistors and all types of integrated circuits

However, their ability to conduct electricity can be greatly improved by adding certain "impurities" to this crystalline structure thereby, producing more free electrons than holes or vice versa.

By controlling the amount of another material added to the semiconductor, it is possible to control its conductivity. These impurities are called donors or acceptors depending on whether they produce electrons or holes respectively. This process of adding impurity atoms to semiconductor atoms (the order of 1 impurity atom per 10 million (or more) atoms of a semiconductor) is called *Doping* and produces an impurity semiconductor.

Figure 2. A Silicon Atom Structure



A single silicon atom shown in Figure 2. consists of fourteen negatively charged electrons surrounding a nucleus of fourteen positively charged protons. Thus silicon has the atomic number 14.

As there are an equal number of positive and negative charges, the silicon atom has no net electric charge. Of the fourteen electrons, only the four outer electrons (shown as those in the outermost valence shell) are available for sharing.

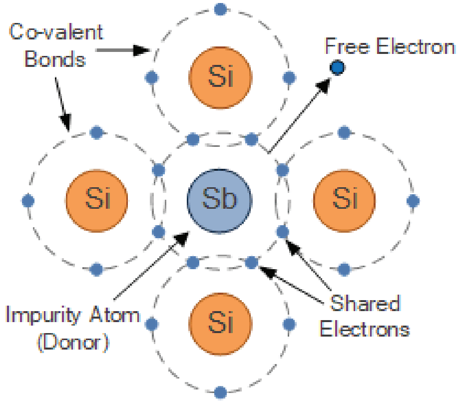
The outer four electrons of a silicon atom bond to other silicon atoms with each bond consisting of two electrons, one electron from each silicon atom. This type of bonding, where the outermost electrons are shared equally between neighbouring atoms is called "covalent bonding".

So in order for a silicon crystal consisting of many such bonds to conduct electricity, we need to introduce an impurity atom that has five outer electrons in its outermost valence shell to share with its neighbouring atoms.

3.1 N-type Semiconductor Structure

The most common type of *Pentavalent* (5-electron) impurity used to dope silicon atoms is Antimony (symbol Sb) or Phosphorus (symbol P).

Figure 3. N-type Atom Structure



These atoms have 51 electrons arranged in five shells around their nucleus with five (5) electrons circling around in its outermost valence shell and therefore one more than the silicon crystal structure. Thus this additional electron can be used for sharing as shown in Figure 3.

After bonding the resulting silicon semiconductor material has an excess of current-carrying free electrons, each with a negative charge.

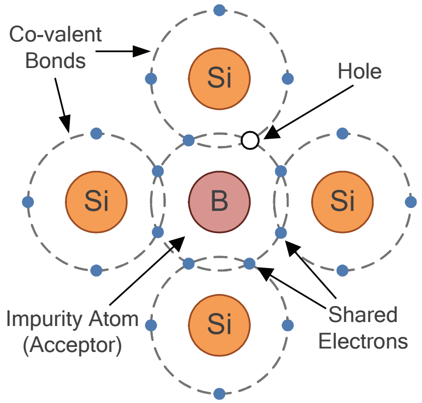
This makes it an n-type, "n" for negative charge carrier, material. These circulating free electrons are called "Majority Carriers" and resulting holes minority carriers.

Then n-type semiconductors have pentavalent impurity atoms (Donors) added to them and conduct by "electron" movement. In these types of semiconductor materials, the donors are positively charged so there will be a large number of "free electrons" which are free to move around from atom to atom. Thus n-type semiconductor materials contain more electrons than holes.

3.2 P-type Semiconductor Structure

If we now go the other way, and introduce a "trivalent" (3-electron) impurity into the crystalline structure, such as Boron (symbol B) or Indium (symbol In), which have only three instead of four outermost valence electrons available for bonding, the fourth complete bond cannot be formed.

FIGURE 4. P-TYPE ATOM STRUCTURE



Therefore because of the missing electron, a complete connection is not possible, giving the semiconductor material an abundance of positively charged carriers. These positive charge carriers are known as "holes".

The doping of Boron atoms causes conduction to consist mainly of positive charge carriers resulting in what is called a "p-type" material with the positive holes being called "Majority Carriers" and any free electrons called minority carriers.

Then p-type semiconductors are a material which have trivalent impurity atoms (Acceptors) added and conducts by the movement of "holes". In these types of materials, the acceptors are negatively charged and there are a large number of holes for free electrons to fill. Thus p-type semiconductor materials contain more holes than electrons.

Then by using different doping agents to a base semiconductor material of either Silicon (S) or Germanium (Ge), it is possible to produce different types of basic semiconductor materials.

A p-type material has positive charge carriers, and the n-type material has negative charge carriers. Thus we could think of semiconductors as being "part-time" conductors whose conductivity can be controlled by the amount of impurity doping.

Donor atoms produce an n-type material with more electrons than holes. Acceptor atoms produce a p-type material with more holes than electrons

4. The PN-junction

The Positive-Negative-Junction, or PN-junction for short, is the basic structure of diodes, rectifiers, transistors, solar cells, light-emitting diodes and any other electronic device which is manufactured using semiconductor materials.

A doped silicon crystal will have either free electrons or stationary holes depending upon whether it's an n-type or p-type material. We now know that an n-type material has a large number of free electrons (being negatively charged) and that the p-type material has a large number of free holes (being positively charged).

As both electrons and holes each have an opposite electrical charge, they will be attracted to each other when brought together. This attraction of opposite charges is called diffusion and is important in understanding the formation of pn- junctions.

When the n-type and p-type materials are first joined together a very large density gradient exists between both sides of the newly created pn-junction. The result is that some of the free electrons from the negative side begin to migrate across this newly formed junction to fill up the holes in the p-type material producing negative ions.

However, because the electrons have moved across the junction from the negative side to the positive side, they leave behind positively charged donor ions on the negative side. This causes holes from the acceptor impurity to migrate across the junction in the opposite direction into the region where there are large numbers of free electrons.

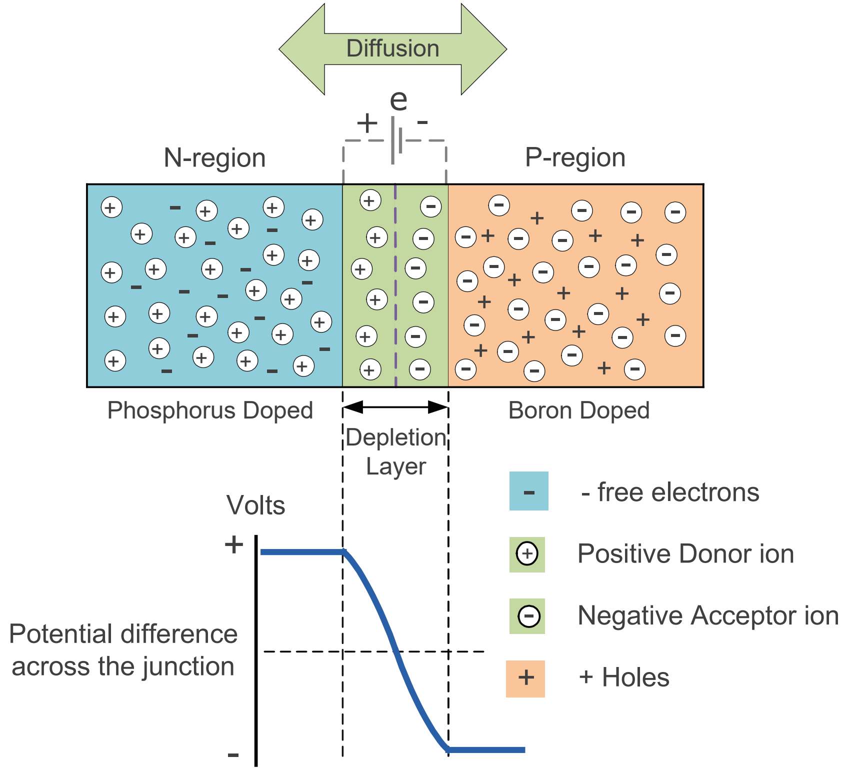
As a result of this movement from one side to the other, the charge density of the p-type along the junction is filled with negatively charged acceptor ions. Likewise, the charge density of the n-type along the junction becomes positive.

That is, free electrons in the n-type region migrate over and fill the holes in the p-type region with this transfer of electrons and holes across the junction being known as *diffusion*.

This process continues back and forth until the number of electrons which have crossed the junction have a large enough electrical charge to repel or prevent any further charge carriers from migrating across the newly formed junction.

Eventually a state of equilibrium (electrically neutral situation) will occur producing a "potential barrier" zone around the area of the junction as the donor atoms repel the holes and the acceptor atoms repel the electrons. For a silicon diode, this happens at around 0.7 volts. Thus diffusion establishes a "built-in" electric field as shown in Figure 5.

Figure 5. Depletion Layer Formation



Since no free charge carriers can rest in a position where there is a potential barrier, the regions on either sides of the junction now become completely "depleted" of any more free carriers in comparison to the n- and p-type materials further away from the junction. As a result, the depletion region becomes highly resistive and behaves as if it were a perfect insulator. The area around the PN-Junction is now called the Depletion Layer.

The width of the p- and n-type areas inside the depletion layer depends on how heavily each side is doped. The total charge on each side of a pn-junction must be equal and opposite to maintain a neutral charge condition around the junction.

Diffusion is the movement of electrons and holes across the pn-junction establishing a built-in electric field around it.

The electric field created by the diffusion process has produced a potential difference across the junction with an open-circuit (zero bias) condition.

The voltage across the depletion layer for silicon is about 0.6 – 0.7 volts and for germanium is about 0.3 – 0.35 volts. This potential barrier will always exist even if the device is not connected to any external power source, as seen in diodes.

The significance of this built-in potential across the junction is that it opposes both the flow of holes and electrons across it and is why it is called the potential barrier. In practice, a pn-junction is formed within a single crystal of semiconductor material rather than just simply joining or fusing together two separate pieces.

5. Biasing The PN-Junction

Without any external voltage being applied to the newly created pn-junction, the junction just sits there in a state of equilibrium. However, if we were to make electrical connections at the ends of both the n-type and the p-type materials and connect them to a battery source, additional energy now exists to overcome the potential barrier.

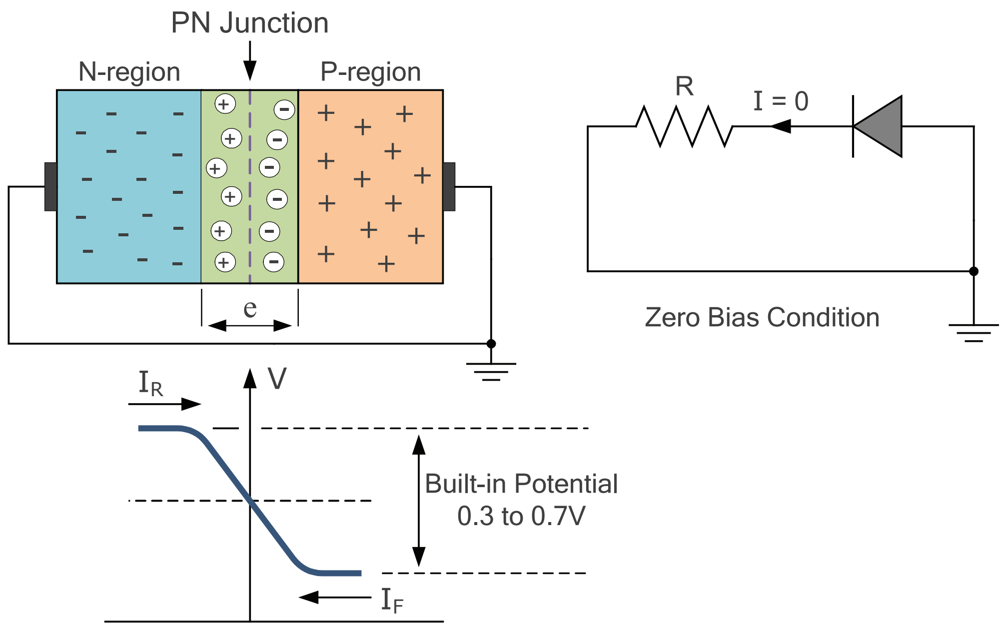
The effect of adding this additional energy source results in the free electrons being able to cross the depletion region from one side to the other. The biasing of the pn-junction with regards to the potential barrier's width produces an asymmetrical conducting two terminal device, better known as the: **pn-Junction Diode**.

There are two operating regions and three possible "biasing" conditions for the standard Junction Diode and these are:

- **Zero Bias** – No external voltage potential is applied to the pn-junction and is in a state of equilibrium.
- **Reverse Bias** – The voltage potential is connected negative, (-ve) to the p-type material and positive, (+ve) to the n-type material across the junction which has the effect of Increasing the pn-junction's width.
- **Forward Bias** – The voltage potential is connected positive, (+ve) to the p-type material and negative, (-ve) to the n-type material across the junction which has the effect of Decreasing the pn-junction's width.

5.1 Zero Biasing the PN-junction

Figure 6. Zero Bias

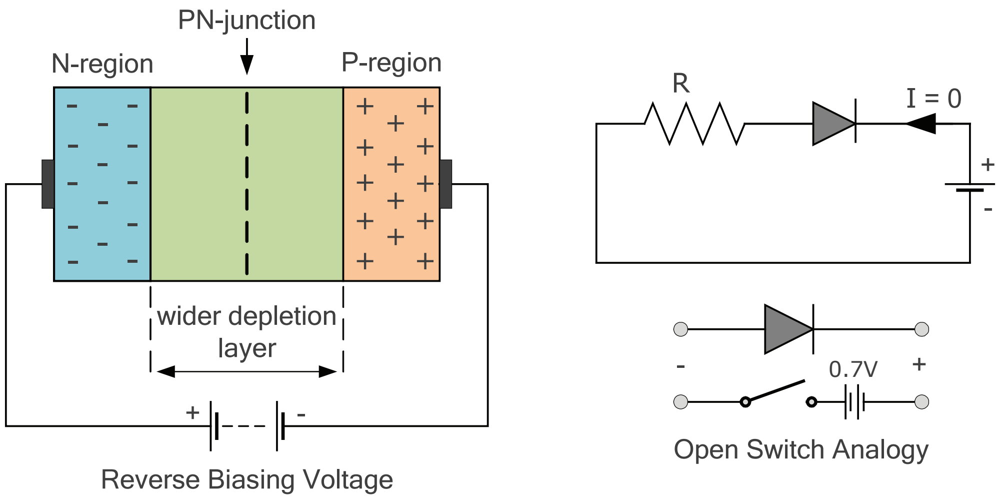


When a semiconductor diode is connected in a *Zero Bias* condition as shown in Figure 6, no external voltage is applied across the pn-junction. If the diodes terminals are shorted together, a few holes in the p-type material with enough energy to overcome the potential barrier of the junction will move across the junction against this barrier potential.

This movement is known as the Forward Current, I_F. Likewise, holes generated in the n-type material move across the junction in the opposite direction. This is known as the Reverse Current, I_R. Although a voltage difference appears across the junction due to the diffusion process, no electric current flows since no external circuit has been connected to the pn-junction.

5.2 Reverse Biasing the PN-junction

Figure 7. Reverse Bias

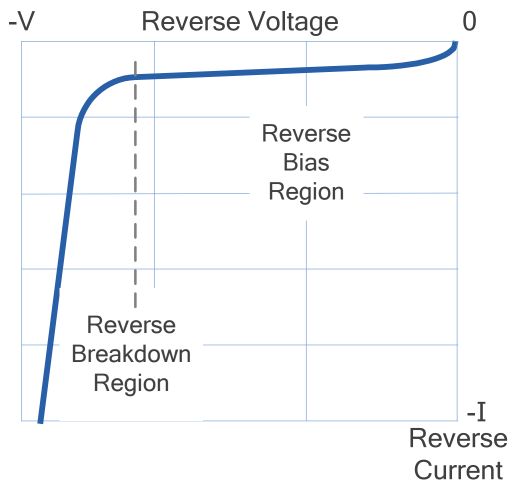


When the pn-junction is connected in a Reverse Bias condition as shown in Figure 7, a positive voltage is applied to the n-type material and a negative voltage is applied to the p-type material.

The positive voltage applied to the n-type material attracts electrons towards the positive electrode and away from the junction, while the holes in the p-type end are also attracted away from the junction towards the negative electrode.

The net result is that the junctions built-in electric field and the reverse external voltage add together as they are both in the same voltage direction causing the depletion layer to grow even wider.

FIGURE 8. REVERSE BREAKDOWN



Due to a lack of electrons and holes in the depletion layer, it presents a high impedance path (an open switch). This prevents current from flowing through the junction. Thus for a reverse bias condition, a pn-junction acts as an open-switch.

However, a very small leakage current does flow through the junction which can be measured in micro-amperes, (μA).

Increasing the external source voltage makes the depletion layer become wider and more resistive until the point where the external voltage is sufficiently high enough to cause permanent failure of the junction as shown on the I-V curve of Figure 8.

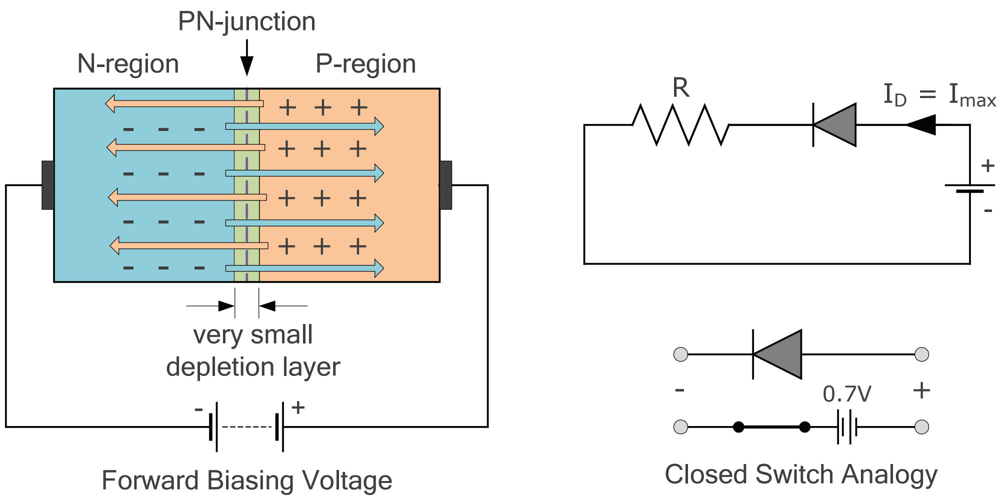
Having said that, the failure of the pn-junction due to an increase in the external reverse voltage has practical applications in many voltage stabilising circuits. It is possible to design the reverse operating characteristics of the pn-junction in such a way that the breakdown voltage is at a specific desired value.

A resistor in series with the diode is used to limit the reverse breakdown current to some preset maximum value thus producing a fixed voltage output across the diode junction. These types of diodes are commonly known as Zener Diodes and are discussed later in chapter.

5.3 Forward Biasing the PN-junction

When the pn-junction is connected in a Forward Bias condition as shown in Figure 9, a negative voltage is applied to the n-type material and a positive voltage is applied to the p-type material. The net result is that the junctions built-in electric field and the reverse external voltage are in opposite directions causing the depletion layer to become thinner in magnitude than the magnitude of the original "built-in" zero biased condition.

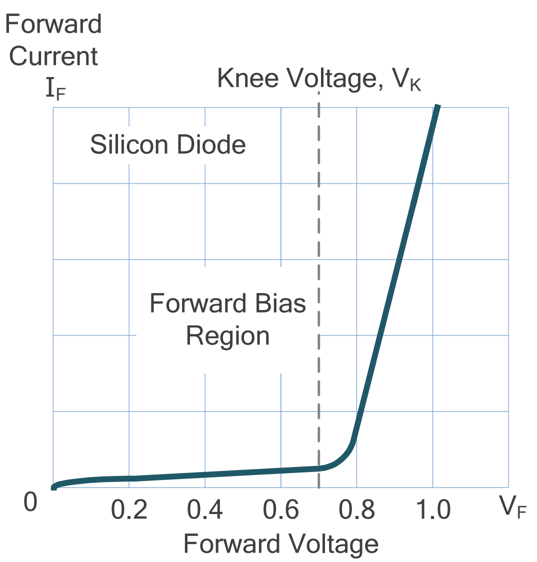
FIGURE 9. FORWARD BIAS



If this external voltage becomes greater than the value of the potential barrier, about 0.7 volts for *silicon semiconductors*, and 0.3 volts for *germanium semiconductors*. The potential barriers opposition will be overcome and electrical current will start to flow unimpeded through the junction.

This is because the negative side of the external voltage pushes or repels electrons towards the junction giving them the energy to cross over and combine with the holes being pushed in the opposite direction towards the junction by the positive side of the external voltage. Thus the potential energy of the power supply is stronger than that within the diode as free electrons will always move towards the strongest force (charge).

FIGURE 10. FORWARD BIAS



This forward bias condition represents the low impedance path (a closed switch) condition through the pn-junction. This allows large currents to flow with only a small increase in the external biasing voltage. Thus for a forward bias condition, a pn-junction acts as a closed-switch.

This results in a forward characteristics curve of Figure 10 having virtually zero current flow up to this voltage point, called the "knee".

Above the 0.7 volt point (for silicon) current flows through the diode with very little increase in the external voltage as shown. In other words, an external voltage of at least 0.7 volts (silicon) must be applied across the diode for it to conduct. For a semiconductor diode, this value is called the barrier potential, or threshold voltage.

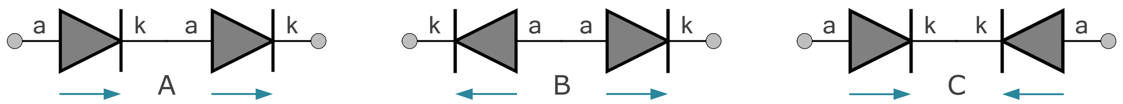
Since the diode can conduct "infinite" current above this knee point as it effectively becomes a closed switch, so resistors are used in series with the diode to limit the current flow to a safe value. Also as long as the pn-junction is conducting current, the forward voltage drop V_F across the diode is always equal to the knee voltage, V_K

6. THE SEMICONDUCTOR DIODE

We have seen that the pn-junction is the basic building block of all semiconductor diodes. Current rises rapidly through the junction when forward biased and practically zero current flows when reverse biased, up to the junctions reverse breakdown voltage. Then we can use this "open" or "closed" switching effect of the junction to good use in many different circuits and applications.

However, as semiconductor diodes are one way switching devices, they cannot be connected together in series randomly. In Figure 11, only circuit combination "A" would conduct or pass an electric current.

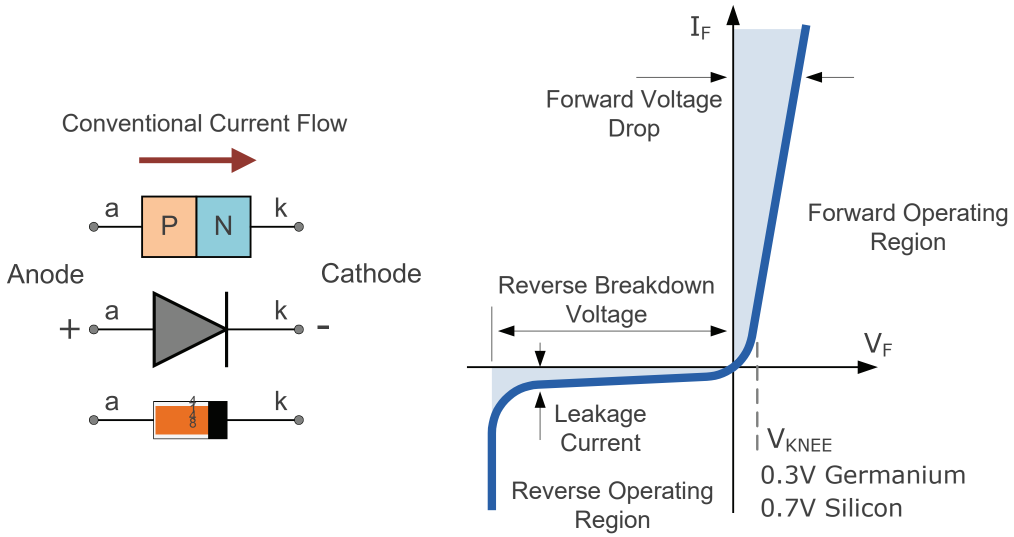
FIGURE 11. CONNECTING DIODES



There are basically two types of pn-junction semiconductor diode, the small *Signal Diode* and the *Power Diode*. The signal diode (for example the 1N4148) is a small semiconductor device generally used in electronic circuits, where small currents or high frequencies are involved such as in clamping, wave-shaping or digital logic circuits.

Power diodes (the 1N4007) on the other hand have much larger semiconductor pn-junctions compared to the smaller signal diode, and are therefore capable of passing much large forward currents at high voltage values for use in rectification circuits.

FIGURE 12. SEMICONDUCTOR DIODE I-V CHARACTERISTICS CURVE



The standard electronic symbol given for any type of diode, either signal or power diode, is that of an arrow with a bar or line at its end and this is illustrated in Figure 12. along with the diodes *Steady State I-V Characteristics Curve*.

The arrow head always points in the direction of conventional current flow through the diode. This means that the diode will only conduct if a positive supply is connected to the Anode, (a) terminal and a negative supply is connected to the Cathode (k) terminal. The cathode (negative end) is often marked on the diodes body with a band for identification.

7. Semiconductor Diode Parameters

PN-junction diodes are manufactured in a wide variety of voltage and current ratings which must be taken into account when choosing a diode for a particular application or circuit. Typical diode parameters to considered are:

- Maximum Forward Current, (I_F)
- Forward Voltage Drop, (V_F)
- Peak Inverse Voltage, (PIV)
- Maximum Operating Temperature, (T)

When forward biased and conducting, a diodes pn-junction has a small forward dynamic resistance, so I^2*R power must be dissipated across it as it conducts causing heat to be generated directly at the junction. Therefore to prevent overheating of the pn-junction, a diode has a **Maximum Forward Current** rating.

The **Forward Voltage Drop** is the voltage drop across anode and cathode terminals of the diode at a defined current level when the diode is forward biased. Commonly it is given by the relationship of: P_D/I_F.

The **Peak Inverse Voltage** is the maximum voltage value allowed across the anode and cathode terminals of a diode in the reverse direction without reverse breakdown occurring. Thus PIV represents the maximum reverse voltage that may be applied to a diode. For a small signal diode (a 1N4148) its PIV maybe as little as 50 volts, while a power diode (1N4007) can withstand a PIV of 1000 volts before breaking down.

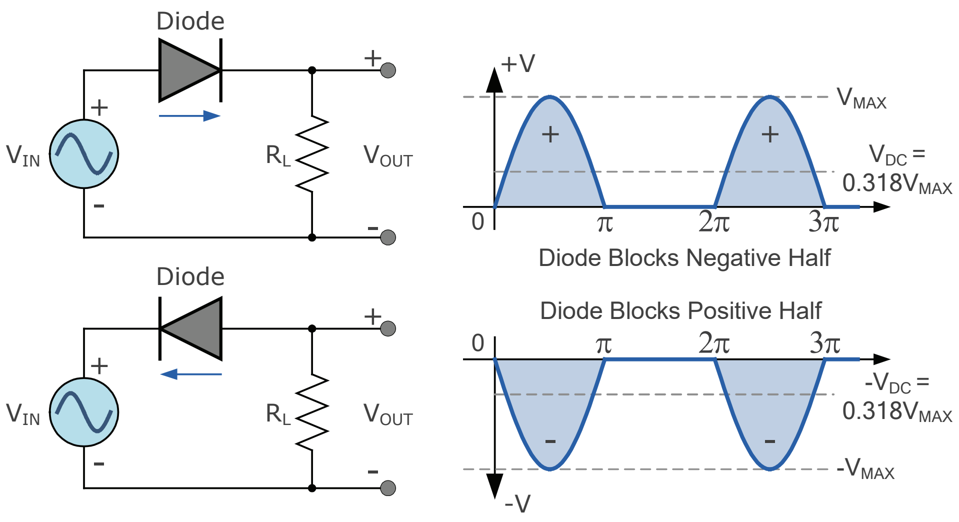
The **Maximum Operating Temperature** is the maximum ambient temperature that the diode can operate in before the pn-junction of the diode deteriorates. The maximum forward current is chosen so that the junction temperature does not exceed this.

8. Half-wave Rectification

One particular application of a pn-junction diode is that of a rectifying diode. When used as a rectifier, either the positive or negative half of an alternating input waveform is blocked, depending on the orientation of the diode. In fact one of the fundamental applications of the diode is in Rectification.

Rectification is the process of changing an alternating AC input supply into a pulsed or continuous DC output supply that has only one single polarity as shown in Figure 13.

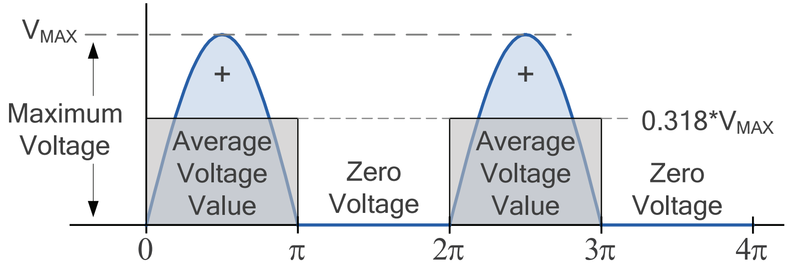
Figure 13. Half-wave Rectification



During each half cycle of the AC sine wave in which the diode is forward biased, that is the Anode is positive with respect to the Cathode, results in current flowing through the diode. Thus: $I_{OUT} = I_{IN}$. During the other half cycle of the AC input waveform, the diode is reverse biased as the Anode is negative with respect to the Cathode. Therefore, no current flows through the diode so $I_{OUT} = 0$

The current on the DC side of the circuit flows in one direction only making the circuit unidirectional. As the load resistor receives from the diode one half of the waveform, then zero volts, then one half of the waveform, then zero volts etc, producing an output which consists of half-cycles only. This pulsating output waveform not only varies ON and OFF every cycle, but is only present 50% of the time as shown in Figure 14.

Figure 14. Half-wave Rectifier Average Output Value



This pulsating voltage means that the equivalent DC value dropped across the load resistor, R_L is therefore only one half of the sinusoidal waveforms average value. So the equivalent DC voltage value taken over one-half of a waveform is defined as being: 0.637 x maximum amplitude value. Thus the equivalent DC voltage is given as: $0.318{*}V_{MAX}$ for the maximum voltage amplitude or $0.45{*}V_{RMS}$ for the root mean squared value of each half-cycle.

$$V_{AVE} = 0.318 \times V_{MAX} = 0.45\ V_{RMS}$$

and

$$I_{AVE} = 0.318 \times I_{MAX} = 0.45\ I_{RMS}$$

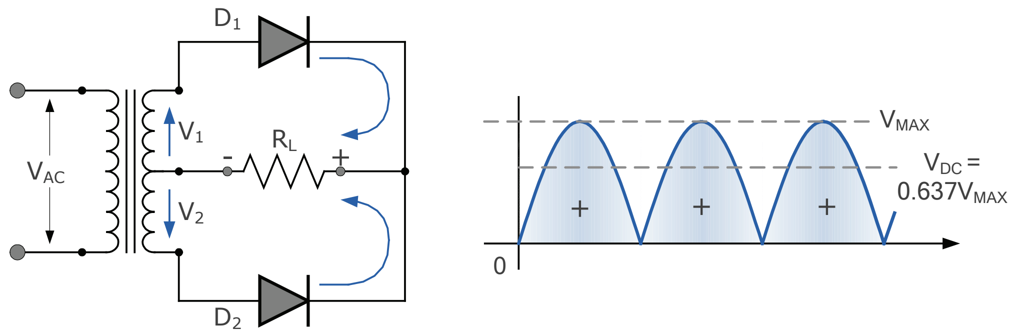
Then a half-wave rectifier circuit converts either the positive or negative halves of an AC waveform into a pulsed DC output that has an average value of $0.318{*}A_{MAX}$ or $0.45{*}A_{RMS}$.

9. Full-wave Rectification

The **full-wave rectifier** utilises both halves of the input sinusoidal waveform to provide a unidirectional output that basically consists of two half-wave rectifiers connected together to feed the connected load.

The full-wave rectifier does this by using two diodes to pass both the negative and positive half cycles of the input waveform, while inverting the negative half of the input waveform to create a pulsating DC output. Even though the voltage and current output from the rectifier is pulsating, it does not reverse direction using the full 100% of the input waveform and thus providing full-wave rectification as shown in Figure 15.

Figure 15. Full-wave Rectification

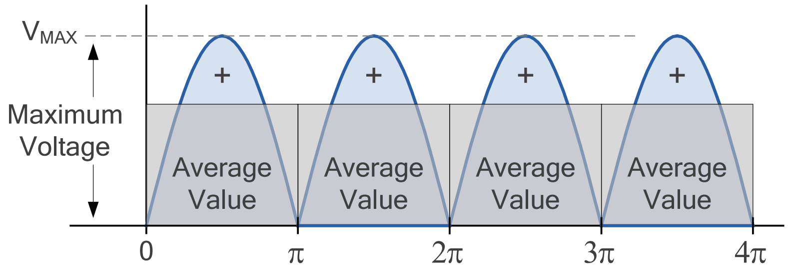


Although this pulsating output waveform uses 100% of the input waveform, its average DC voltage (or current) is not at the same value as the half-wave rectifier.

We remember from the half-wave rectifier configuration that the average or mean DC value taken over one-half of a sinusoid is defined as: 0.318 x maximum amplitude value.

However unlike half-wave rectification, full-wave rectifiers have two positive half-cycles per input waveform giving us a different average value as shown.

Figure 16. Full-wave Rectifier Average Output Value



As with the previous half-wave rectifier output, for each positive peak there is an average DC value of $0.318*V_{MAX}$. But for a full-wave rectifier output, there are two peaks per input waveform. This means there are two lots of average value per cycle of input waveform as shown in Figure 16.

Thus the DC output voltage of a full-wave rectifier will therefore be twice that of the previous half-wave rectifier giving an average value of $0.637*V_{MAX}$ ($2 \times 0.318V_{MAX}$).

Then the equivalent DC voltage for an unsmoothed full-wave rectifier is therefore given as: $0.637*V_{MAX}$ for the maximum voltage amplitude, or $0.9*V_{RMS}$ for the root mean squared value of each cycle. Thus:

$$V_{AVE} = 0.637 \times V_{MAX} = 0.9\ V_{RMS}$$

and

$$V_{AVE} = 0.637 \times V_{MAX} = 0.9\ V_{RMS}$$

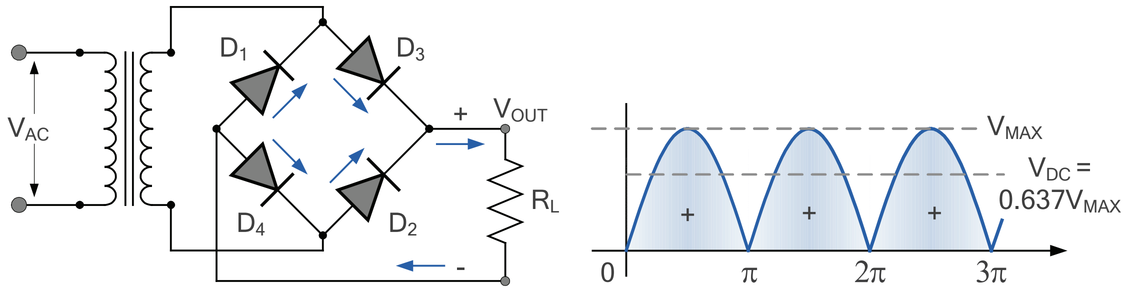
Then we can see that a full-wave rectifier circuit converts both the positive or negative halves of an AC input waveform into a pulsed DC output as there are two diode paths for current to follow. So has an average value of $0.637*A_{MAX}$ or $0.9*A_{RMS}$.

Note that a full-wave rectifier circuit also reduces the output ripple by a factor of two, but the output DC ripple is twice the input supply frequency.

10. The Bridge Rectifier

The **full-wave bridge rectifier** circuit uses four diodes arranged in a bridge arrangement to pass both halves of the input waveform to the output producing a pulsating DC output. At any time two of the four diodes in the bridge are forward biased while the other two are reverse biased. Thus there are two diodes in the conduction path instead of the single one for the half-wave rectifier as shown in Figure 17.

Figure 17. Bridge Rectifier Circuit



During the positive half cycle of V_{AC}, diode pair D_3 and D_4 are forward biased and conducting while diode pair D_1 and D_2 are reverse biased and OFF. Then for the positive half cycle of the input waveform, current flows along the path of: $D_3 – R_L – D_4$ and back to the supply.

During the negative half cycle of V_{AC}, diode pair D_2 and D_1 are forward biased and conducting while diode pair D_3 and D_4 are reverse biased and OFF. Then for the negative half cycle of the input waveform, current flows along the path of: $D_2 – R_L – D_1$ and back to the supply.

Thus, the four diodes conduct in pairs of two and depending upon the polarity of the input waveform determines which pair conduct and the load will always see the same voltage polarity across itself.

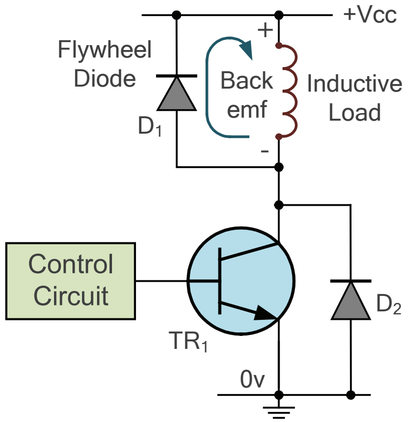
While we can use four individual diodes to make a full-wave bridge rectifier, ready-made bridge rectifier are available “off-the-shelf” in a range of different voltage and current sizes.

11. The Flywheel Diode

When a transistor, MOSFET, switch or sensor is used to control an inductive load, for example a relay coil, solenoid, winding, small DC motor, etc. the electrical energy stored within the magnetic field of the load will subject any switching device to a high reverse voltage when the load current is quickly disconnected (turned-off).

This is because opening of the switching device does not immediately stop current flowing in the inductive load. Thus we need a way to safely dissipate the energy stored in the inductance of the load when the switching device is opened and we can do this by using a voltage transient suppression diode.

Figure 18. Flywheel Diode Protection



Transient voltage suppression diodes are designed to protect fragile semiconductor components from large transient spikes that are caused by inductive loads. These diodes are commonly called **flywheel diodes**, **flyback diodes**, or **freewheeling diodes**.

As shown in Figure 18, the diode is connected in parallel across the inductive load, shown by D_1 or across the semiconductor switching device as shown by D_2, or both.

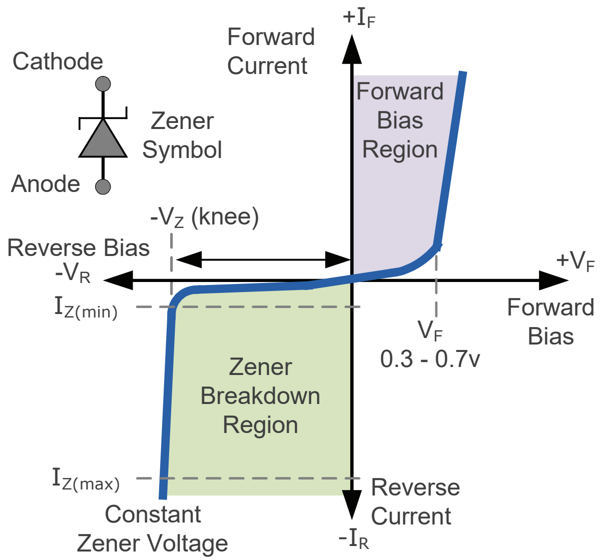
When the inductive load is switched-ON and passing current, diode D_1 is reverse-biased and blocking. But when the load is switched-OFF by TR_1, the rapid collapse of the coils magnetic field causes a large back-emf of possibly hundreds of volts to build up across the coil.

As this back-emf is the reversal of the original DC supply, the diode now becomes forward biased and clamps the reverse voltage across the coil to about 0.7V dissipating the stored energy and protecting the switching transistor. Flywheel diodes are only applicable when the supply is a polarised DC voltage. Transient suppression of an AC coil requires a different protection method, and for this an RC snubber network or MOV is used.

12. The Zener Diode

Previously we said that if a normal pn-junction diode is reverse-biased with a sufficiently large enough reverse voltage, it will suffer from premature breakdown and become damaged. The **Zener Diode** or "*Breakdown Diode*", as they are sometimes referred too, is a type of semiconductor diode which is designed to take advantage of this reverse voltage breakdown condition.

Figure 19. Zener Diode I-V Characteristics



A zener diode behaves just like a normal diode when biased in the forward direction passing the rated forward current.

However, unlike a normal pn-junction diode which blocks any flow of current through itself when reverse biased, as soon as the reverse voltage reaches a pre-determined value of the zener diode, it begins to conduct current in the reverse direction.

A Zener Diode is used in its "reverse bias" or reverse breakdown mode for voltage regulation in that the diodes anode connects to the negative side of the supply. When sufficient reverse voltage is applied (cathode end biased positively), the zener is driven into its reverse breakdown avalanche mode of operation.

From the I-V characteristics curve of Figure 19. we can see that the transition into avalanche breakdown, the "knee" point, a zener diode has an almost constant negative voltage, V_Z regardless of the value of the current flowing through the diode.

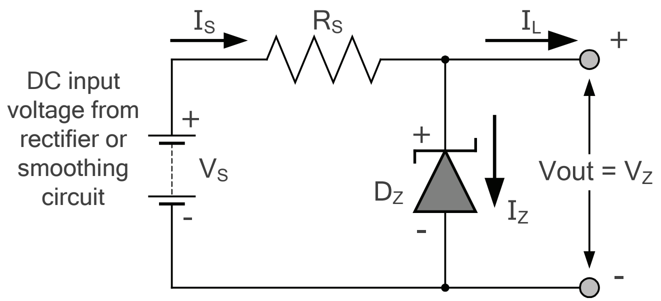
This zener value remains nearly constant even with large variations in forward current as long as the zener diodes current remains between its breakdown current, $I_{Z(MIN)}$ and its maximum handling current rating, $I_{Z(MAX)}$.

Thus a zener diode can conduct current in both directions with the forward current I_F being a function of forward voltage V_F, and the reverse zener current, I_Z being a function of the reverse breakdown voltage V_Z.

13. Zener Diode Regulation

The ability of a zener diode to control itself can be used to great effect regulating or stabilising a voltage source against supply or load variations. The fact that the voltage across a zener diode in the breakdown region is almost constant turns out to be an important characteristic as voltage regulation is a common application of a zener diode.

Figure 20. Zener Diode Voltage Regulator



The basic voltage regulator circuit of Figure 20. shows that resistor, R_S is connected in series with the zener diode to limit the current flowing through the diode. The supply voltage, V_S which is greater than V_Z is connected across the series combination.

The stabilised output voltage V_{OUT} which is also the zener voltage V_Z is taken from across the zener diode.

The zener diode is connected with its cathode terminal connected to the positive rail of the DC supply so it is reverse biased and will be operating in its breakdown condition. Resistor R_S is selected so to limit the maximum current flowing in the circuit and power dissipation in a zener diode.

With no load connected to the circuit, the load current will therefore be zero, ($I_L = 0$), so all the circuit current passes through the zener diode which in turn dissipates its maximum power. The load resistance R_L is connected in parallel with the zener diode thus $V_{OUT} = V_Z$. The supply current is now split between the zener diode and the load with the voltage drop across the series resistor being: $V_S - V_Z$.

Under light load (high-impedance) conditions, nearly all the supply current flows through the zener diode and so the zener diode will attempt to limit the output voltage to the zener potential. Under heavier load (low-impedance) conditions, most of the supply current is taken by the load with less current flowing through the zener diode.

If the current drawn by the load current becomes too excessive, then there is no supply current available for the zener diode so it stops conducting and so output voltage regulation is lost.

Zener Diodes can be used to produce a stabilised voltage output

Then there is a minimum zener current, I_Z for which the stabilization of the zener voltage, V_Z is effective so the zener current must stay above this value operating under load within its breakdown region at all times. The upper limit of current is of course dependent upon the power rating of the zener device

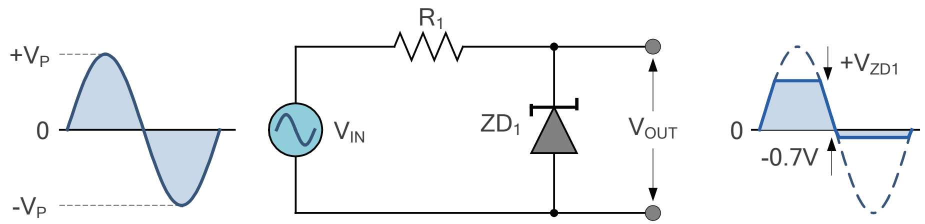
14. Zener Diode Clipping Circuits

As well as voltage regulation circuits, zener diodes can also be used to limit the maximum amplitude of a signal. Zener diode clipping and clamping circuits can shape or modify an input AC waveform (or any sinusoid) producing a differently shape output waveform depending upon the circuit arrangement.

Diode clipping circuits, also called limiters because they limit or clip-off the positive (or negative) part of an input AC signal waveform are mainly used for circuit protection or for use in waveform shaping circuits. For example, if we wanted to clip a sinusoidal waveform at +7.5 volts, we would use a 7.5V zener diode. If the signal waveform exceeds this 7.5V limit, the zener diode will “clip-off” the excess voltage from the input producing a waveform with a flat top keeping the output constant at the required +7.5 volts.

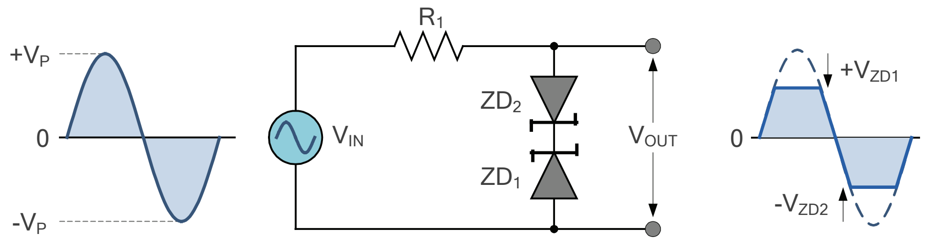
Note that in the forward bias condition a zener diode is still an ordinary semiconductor diode and when the AC waveform goes negative below -0.7V, (for silicon) the zener diode turns "ON" (conducts) like any normal silicon pn-junction diode would and clips the output at -0.7V as shown in Figure 21. producing a half-wave clipping circuit.

Figure 21. Half-wave Zener Diode Clipping



We can develop this idea further by using the zener diodes reverse-voltage characteristics of two zener's to clip both halves of a signal waveform using series connected back-to-back zener diodes as shown in Figure 22.

Figure 22. Full-wave Zener Diode Clipping



The output waveform from a full-wave zener diode clipping circuit will be clipped at the zener voltage plus the 0.7 volt (silicon) forward volt drop of the other diode. So for example, the positive half-cycle will be clipped at the sum of zener diode, ZD_1 plus the forward-biased 0.7 volts from ZD_2 and vice versa for the negative half-cycle.

Zener diode clipping circuits can be used to limit the maximum amplitude of a signal waveform. They can clip only the positive part of the signal waveform, the negative part, or both polarities producing an almost square-wave waveform signal from a sinusoid.

The positive and negative clipping levels can be adjusted independently by using two back-to-back zener diodes with different zener voltage clamping ratings. For example, the positive half-cycle could be at 4.5 volts and the negative half-cycle at 7.2 volts.

Zener diodes are manufactured with a wide range of voltages and can be used to give different voltage references on each half cycle, the same as above. Zener diodes are available with nominal zener breakdown voltages, V_Z ranging from 2.4 to over 100 volts, (for example the 1N4728A to 1N4764A range) with a typical tolerance of 1 or 5%. Note that once conducting in the reverse breakdown region, full current will flow through the zener diode so a suitable current limiting resistor, R_Z must be chosen.

End of this Semiconductor Diode eBook

With the completion of this Semiconductor Diode eBook you should have gained a basic understanding and knowledge of diodes. The information provided here should give you a firm foundation for continuing your study of electronics and electrical engineering. In ebook 12 you will learn about using power diodes for rectification.

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